Name: Atharv Revat Shukla

PRN:25070122053

Division: A2

Subject: Programming with Java

ASSIGNMENT No : 11

TITLE :Write a Java program to demonstrate the implementation of interfaces.

Problem Statement

Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives

To implement interfaces for achieving abstraction and multiple inheritance of type.

Theory

An interface specifies a set of abstract methods that implementing classes must define. It provides

complete abstraction and supports multiple inheritance of type. Interfaces improve modularity, loose

coupling, and software flexibility. They allow different classes to implement common behavior

while maintaining independent implementations. Interfaces are widely used in enterprise application

development.

Outcome

Conclusion

These programs demonstrate the use of **interfaces**, where different classes implement the same methods in their own way.

Exercise

1. Create an interface Printable and implement it in Student and Employee classes.

```
import java.util.Scanner;
```

```java
interface Printable {
    void print();
}
class Student implements Printable {
    String name;

    Student(String name) {
        this.name = name;
    }
    public void print() {
        System.out.println("Student Name: " + name);
    }
}
class Employee implements Printable {
    String name;
    Employee(String name) {
        this.name = name;
    }
    public void print() {
        System.out.println("Employee Name: " + name);
    }
}
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Student Name: ");
        String s = sc.nextLine();
        System.out.print("Enter Employee Name: ");
        String e = sc.nextLine();
```

```java
        Student st = new Student(s);
        Employee emp = new Employee(e);
        st.print();
        emp.print();
    }
}
```

2. Create an interface Switchable with a method turnOn(). Implement the interface in Light

and Fan classes to display the device status.

```java
import java.util.Scanner;
interface Switchable {
    void turnOn();
}
class Light implements Switchable {
    public void turnOn() {
        System.out.println("Light is ON");
    }
}
class Fan implements Switchable {
    public void turnOn() {
        System.out.println("Fan is ON");
    }
}
public class Main {
    public static void main(String[] args) {
        Light light = new Light();
        Fan fan = new Fan();
        light.turnOn();
        fan.turnOn();
```

```
    }
}
```